

Mathematical Analysis 2

Seminar 13

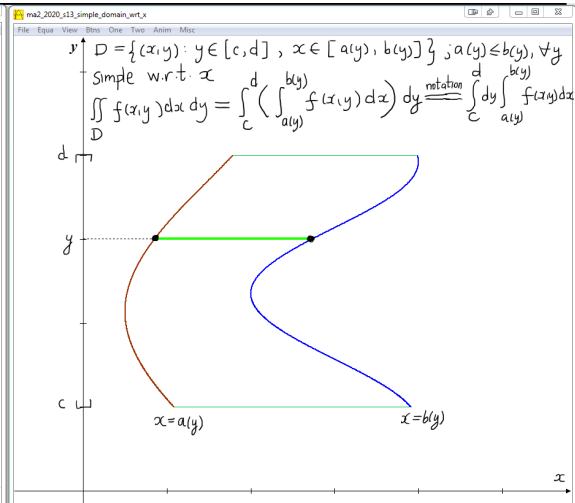
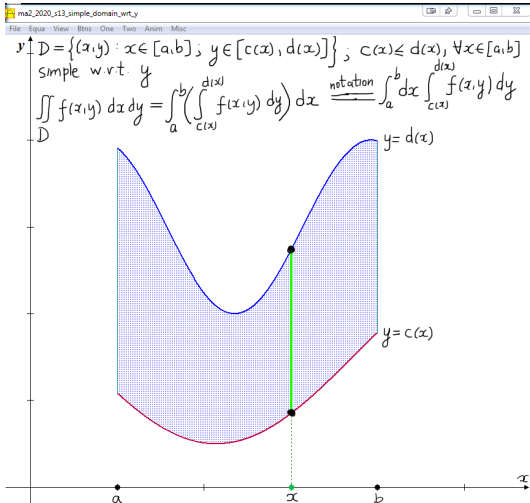
Double integrals (2)

The double integral on rectangular domains (from Seminar 12)

$$\begin{aligned} D = [a, b] \times [c, d] ; f : D \rightarrow \mathbb{R} \text{ continuous} \\ \iint_D f(x, y) \, dx \, dy \quad \xrightarrow{\text{notation}} \quad \int_a^b \int_c^d f(x, y) \, dx \, dy \quad \xrightarrow{\text{computation}} \quad \begin{cases} \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx & \xrightarrow{\text{notation}} \int_a^b dx \int_c^d f(x, y) \, dy \\ \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy & \xrightarrow{\text{notation}} \int_c^d dy \int_a^b f(x, y) \, dx \end{cases} \end{aligned}$$

↗ iterated integrals

The double integral on simple domains w.r.t. the axes of coordinates



Ex. 1

$$\iint_D \frac{x^2}{y^2} dx dy, \quad D \subseteq \mathbb{R}^2 \text{ bounded by } x=2, x=y, xy=1.$$

Intersection points between curves

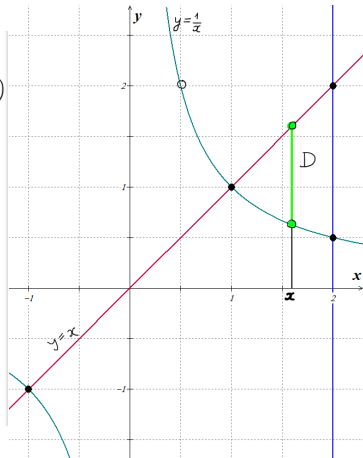
$$\begin{cases} x=2 \\ x=y \end{cases} \rightarrow (2,2); \quad \begin{cases} x=2 \\ xy=1 \end{cases} \rightarrow (2, \frac{1}{2}); \quad \begin{cases} x=y \\ xy=1 \end{cases} \rightarrow (-1,-1) ; (1,1)$$

$$D: \begin{cases} x \in [1, 2] \\ y \in [\frac{1}{x}, x] \end{cases}$$

$$I = \int_1^2 dx \underbrace{\int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy}_{I_1}$$

$$\begin{aligned} I_1 &= x^2 \int_{\frac{1}{x}}^x \frac{1}{y^2} dy = -x^2 \cdot \frac{1}{y} \Big|_{y=\frac{1}{x}}^{y=x} = \\ &= -x^2 \left(\frac{1}{x} - x \right) = -x + x^3 \end{aligned}$$

$$I = \int_1^2 (-x + x^3) dx = \left(-\frac{x^2}{2} + \frac{x^4}{4} \right) \Big|_1^2 = \dots$$



Ex. 2

$$\iint_D \frac{y^3 \cdot \sin \pi x}{x} dx dy,$$

$$D: \begin{cases} y \in [0, 1] \\ y^4 \leq x \leq y^2 \end{cases}$$

simple w.r.t. x

$$I = \int_0^1 dy \int_{y^4}^{y^2} \frac{y^3 \sin \pi x}{x} dx \quad ; \quad \text{change the order of integrals}$$

$I = \int_{\text{?}}^{\text{?}} dx \int_{\text{?}}^{\text{?}} \frac{y^3 \cdot \sin \pi x}{x} dy$

complicated

$$\begin{cases} 0 \leq y \leq 1 \\ y^4 \leq x \leq y^2 \end{cases} \iff \begin{cases} x \in [0, 1] \\ |y| \leq \sqrt[4]{x} ; |y| \geq \sqrt{x}, 0 \leq y \leq 1 \end{cases}$$

($y=0 \Rightarrow x=0$) ($y=1 \Rightarrow x=1$)

$$\iff \begin{cases} x \in [0, 1] \\ y \in [\sqrt{x}, \sqrt[4]{x}] \end{cases} \quad (D \text{ is simple w.r.t } y)$$

$$I = \int_0^1 dx \int_{\sqrt{x}}^{\sqrt[4]{x}} \frac{y^3 \cdot \sin \pi x}{x} dy = \int_0^1 \frac{\sin \pi x}{x} \cdot \frac{y^4}{4} \Big|_{\sqrt{x}}^{\sqrt[4]{x}} dx = \frac{1}{4} \int_0^1 \frac{\sin \pi x}{x} (x - x^2) dx =$$

$$= \frac{1}{4} \int_0^1 (1-x) \cdot \sin \pi x dx \xrightarrow{\text{by parts}} \dots = \frac{1}{4\pi}$$

↑ self study.

Ex. 3

$$\iint_D e^{y^3} dx dy, \quad D: \begin{cases} x \in [0, 1] \\ \sqrt{x} \leq y \leq 1 \end{cases}$$

$$I = \int_0^1 dx \int_{\sqrt{x}}^1 e^{y^3} dy \quad ; \text{ change the order of integrals.}$$

difficult

$$D: \begin{cases} 0 \leq x \leq 1 \\ \sqrt{x} \leq y \leq 1 \end{cases} \Leftrightarrow \begin{cases} y \in [0, 1] \\ x \leq y^2; \underline{x \geq 0}; \underline{x \leq 1} \end{cases} \quad (x=0 \Rightarrow y \in [0, 1]) \Leftrightarrow \begin{cases} y \in [0, 1] \\ x \in [0, y^2] \end{cases}$$

$$I = \int_0^1 dy \int_0^{y^2} e^{y^3} dx = \int_0^1 e^{y^3} \cdot y^2 dy = \left. \frac{e^{y^3}}{3} \right|_0^1 = \frac{e-1}{3}.$$

The change of variable in the double integral

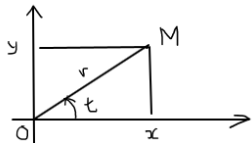
$$\begin{cases} x = x(u,v) \\ y = y(u,v) \end{cases} : (u,v) \in E \longmapsto (x,y) \in D \Rightarrow \iint_D f(x,y) dx dy = \iint_E f(x(u,v), y(u,v)) \cdot |J(u,v)| du dv$$

$$J(u,v) = \det \begin{pmatrix} x'_u & x'_v \\ y'_u & y'_v \end{pmatrix} = \frac{1}{\det \begin{pmatrix} u'_x & u'_y \\ v'_x & v'_y \end{pmatrix}}$$

Polar coordinates (r, t)

$$\begin{cases} x = r \cdot \cos t \\ y = r \cdot \sin t \end{cases} ; r \geq 0, t \in [0, 2\pi) \text{ (or } t \in [-\pi, \pi)) \quad \begin{cases} r^2 = x^2 + y^2 \\ \tan t = \frac{y}{x} \end{cases}$$

$$J(r,t) = \det \begin{pmatrix} \cos t & -r \cdot \sin t \\ \sin t & r \cdot \cos t \end{pmatrix} = r$$

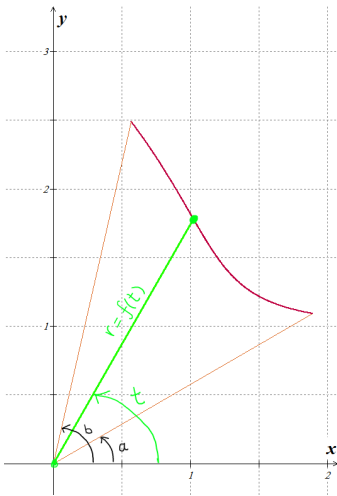


Ex. 4 the area of a disk with radius R

$$D: x^2 + y^2 \leq R^2 \xrightarrow[\substack{\text{polar} \\ (r,t)}]{} E: \begin{cases} r \leq R \\ r \geq 0, t \in [0, 2\pi) \end{cases} = \underbrace{[0, R]}_r \times \underbrace{[0, 2\pi)}_t$$

$$\sigma(D) = \iint_D dx dy = \iint_E \underbrace{r dr dt}_{J(r,t)} = \int_0^R \int_0^{2\pi} r dr dt = \left(\int_0^R r dr \right) \left(\int_0^{2\pi} dt \right) = \frac{r^2}{2} \Big|_0^R \cdot t \Big|_0^{2\pi} = \pi R^2.$$

Ex. 5 (in general) the area in polar coordinates



$$D \xrightarrow{\text{polar}} E : \begin{cases} t \in [a, b] \\ r \in [0, f(t)] \end{cases}$$

$$\begin{aligned} \sigma(D) &= \iint_D dx dy = \iint_E r dr dt = \int_a^b dt \int_0^{f(t)} r dr = \int_a^b \frac{r^2}{2} \Big|_{r=0}^{r=f(t)} dt \\ &= \frac{1}{2} \int_a^b f^2(t) dt \end{aligned}$$

Ⓐ (the cardioid) $f(t) = 2a(1 + \cos t)$, $t \in [-\pi, \pi]$ ($a > 0$)

$$\begin{aligned} \sigma &= \frac{1}{2} \int_{-\pi}^{\pi} 4a^2 (1 + 2 \cos t + \cos^2 t) dt = 2a^2 \left(t \Big|_{-\pi}^{\pi} + 2 \sin t \Big|_{-\pi}^{\pi} + \right. \\ &\quad \left. + \int_{-\pi}^{\pi} \frac{1 + \cos 2t}{2} dt \right) = 2a^2 \left(2\pi + 0 + \frac{1}{2} \left(t + \frac{\sin 2t}{2} \right) \Big|_{-\pi}^{\pi} \right) = \\ &= 2a^2 \left(2\pi + \frac{1}{2} \cdot 2\pi \right) = 6\pi a^2. \end{aligned}$$

Ⓑ (the lemniscate of Bernoulli) $f(t) = \sqrt{\cos t \cdot \sin t}$; $t \in [0, \frac{\pi}{2}]$

$$\sigma = \frac{1}{2} \int_0^{\pi/2} \cos t \cdot \sin t dt = \frac{1}{4} \sin^2 t \Big|_0^{\pi/2} = \frac{1}{4}.$$

Ex. 6

$$\iint_D (x^2 + y^2) dx dy,$$

$$D: x^2 + y^2 \leq 2ax \quad (a > 0 \text{ fixed})$$

$$D \xrightarrow{\text{polar}} E: \begin{cases} r \geq 0, t \in [-\pi, \pi] \\ r \leq 2a \cdot \cos t; \cos t \geq 0 \end{cases}$$

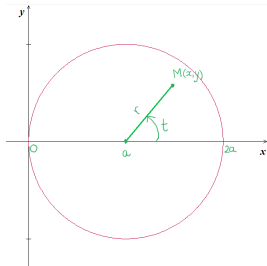
$$0 \leq \underbrace{\frac{r^2}{r} \leq \frac{2a \cdot r \cdot \cos t}{2a \cdot \cos t}}_+$$

$$\begin{aligned} E: \begin{cases} t \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ 0 \leq r \leq 2a \cdot \cos t \end{cases} &\Rightarrow I = \iint_E \underbrace{r^2 \cdot r dr dt}_{dx dy} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dt \int_0^{2a \cdot \cos t} r^3 dr = \frac{1}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \underbrace{(2a \cdot \cos t)^4}_{\text{even}} dt = \\ &= 4a^4 \cdot 2 \int_0^{\pi/2} \cos^4 t dt = 4a^4 \cdot B\left(\frac{1}{2}; \frac{5}{2}\right) = 4a^4 \cdot \frac{\Gamma(\frac{1}{2}) \Gamma(\frac{5}{2})}{\Gamma(3)} = 4a^4 \cdot \frac{\sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{2!} = \\ &= \frac{3\pi a^4}{2} \end{aligned}$$

$$\begin{cases} 2q-1=0 \\ 2q-1=4 \end{cases} B(p,q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cdot \cos^{2q-1} \theta d\theta$$

2nd solution

$$D: (x-a)^2 + y^2 \leq a^2$$

It can be used $(x,y) \rightarrow (r,t)$

$$\begin{cases} x-a = r \cdot \cos t \\ y = r \cdot \sin t \end{cases}; (x-a)^2 + y^2 = r^2$$

$$r \in [0, a], t \in [0, 2\pi]$$

$$E = [0, a] \times [0, 2\pi]$$

$$x^2 + y^2 = r^2 - 2ar \cos t + a^2$$

$$J = r$$

Transforming (almost) everything into rectangles

(next
time)